A Novel DCNN Based MI-EEG Classification Method Using Spatio-Frequency Information

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***Abstract*—Deep convolutional neural network (DCNN) has been successfully applied to improve the classification performance of motor imagery (MI) tasks in the electroencephalogram (EEG) based brain computer interface (BCI). However, there is still a great challenge about how to extract effective features from EEG signals. So, a novel parallel DCNN structure-based MI-EEG classification method is proposed in this paper. Fast Fourier transform is utilized to transform EEG signals from time domain to frequency domain. Then, μ and β rhythms, which are related to MI tasks, are redivided into three sub-bands. The averaged power is calculated for each sub-band. In order to utilize the position information of electrodes, azimuthal equidistant projection is adopted to convert the 3-D location of each electrode into 2-D position. Then, Clough-Tocher interpolation algorithm is employed to generate a spatio-frequency image for each sub-band. Furthermore, a parallel DCNN (pDCNN), which includes three DCNNs correspondings to three sub-bands respectively, is designed to extract and classify spatio-frequency features. The BCI2000 dataset is adopted, and the average accuracy and Kappa value reach 90.25% and 0.81 respectively, which are greater than that of the existing methods. These experimental results show that the effectiveness of the proposed MI-EEG classification method.**

***Index Terms*—Brain computer interface, motor imagery, deep convolutional neural network**

I. Introduction

Brain-computer interface (BCI) provides an effective information exchange way between the brain and the physical devices, which assists people only imaging in their brains to control machines (e.g., BCI-based wheelchairs, upper or lower limb assistive robots) [1]. Due to inexpensive and portable, electroencephalography (EEG) becomes the most common noninvasive method to record brain activity, which has been commonly applied as the input of the BCI system [2]. Nowadays, recognition of motor imagery (MI) (i.e., imagining body movement without actual movement) based on EEG is an outstanding study topic in the BCI fields, which is the focus of this paper.

MI process causes brain activity especially on the motor cortes and induces EEG main change in 8-30Hz band usually divided into μ and β rhythms [3]. However, EEG signals usually perform the characteristic of non-stationary and low signal-to-noise ratio, which affects the classification performance of MI-EEG tasks. Hence, the recognition of the MI-EEG task mainly lies in efficient feature extraction, that is, taking full advantage of the temporal, frequency and space information of the signals. The traditional classification algorithms based on machine learning usually depend on manual extracted features to classify different MI tasks. In [4], a common spatial pattern (CSP) algorithm was used for feature extraction to complete two classification tasks. The support vector machine (SVM) [5], the filter bank common spatial patterns (FBCSP) [6] and other neural network algorithms [7]–[9] were employed for MI classification tasks. However, the effect of manually extracted features directly affects the classification performance, and further the corresponding feature extraction process is usually very complex and requires a significant investment. Hence, how to extract effective features with low cost is an important and intractable problem.

In recent years, due to integrating feature extraction and classification, deep learning (DL) completes end-to-end learning for a specific task, which has been given considerable attention in many fields, e.g., audio and visual recognition. Now, DL based EEG classification approaches have become a hot research topic [10]. A parallel and cascaded neural network based on convolutional neural network (CNN) and long-short term memory (LSTM) was proposed in [11] to directly deal with the original EEG signals. In [12], Fast Fourier transform (FFT) was used to extract the spectrum of MI-EEG, and then a modified VGG16 structure was proposed to achieve the decoding of the MI-EEG tasks. In [13], Stockwell transform (ST) algorithm was adopted to complete time-domain to frequency-domain conversion, and then based on deep metric learning, a

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triplet network framework was constructed to classify MI-EEG tasks. In [14], a CNN hybrid convolution scale model was proposed to solve the problem of low recognition accuracy. The study from [15] proposed an EEG signals classification method based on end-to-end deep learning. And in [16], a QNet, which includes an attention mechanism, was proposed to make the network automatically learn the importance of different electrodes, time points, and feature maps. In summary, DL can avoid the complex feature extraction process usually existing in the traditional classification methods. However, in most existing works, the μ and β rhythms spectrum feature is normally fused before being feeded into CNN network, which may exist the insufficient features extraction, and further cause classification performance degradation. This drawback motivates this study.

In this paper, FFT is adopted to obtain the EEG spectrum, and then three sub-bands are divided from the μ and β rhythms. Based on the average power of each sub-band, Clough-Tocher (CT) interpolation algorithm [17] is applied to obtain a spatio-frequency image for preserving the exact location and frequency information of electrodes. To overcome the insufficient feature extraction, a pDCNN structure including three DCNNs corresponding to the three sub-bands respectively is constructed to classify MI-EEG. This network structure can extract more features from spatio-frequency images, to improve the accuracy of MI-EEG classification tasks.

The rest of this paper is organized as follows. In Section II, the EEG data process and pDCNN structure are detailed. Section III details experiments and the corresponding results. Section IV concludes this paper.

II. METHODS

This section mainly focuses on detailing the overall architecture of the proposed methods for MI-EEG classification tasks from experiment dataset description, EEG data process and the proposed network model, each of which is described in detail as following.

A. BCI2000 Dataset

BCI2000 dataset is recorded by using the BCI2000 system with the sampling rate 160 Hz and includes 64 electrodes. 109 subjects are participating in the MI experiment to record EEG data. And each subject performs 14 experimental runs, including two one-minute baseline runs (i.e., eyes opened and closed), followed by 4 different runs repeated three times, each of which lasts for two minutes. Each subject is required to do one of the following four tasks which are detailed in [18].

In this paper, for this dataset, only the data of imaging open and close left or right fist are utilized for the experiment, which includes 4761 trials. Each trail includes 5 seconds records (i.e., 800 samples) in which the first second corresponds to the rest used to eliminate the baseline [19]. It is well known that two main bands of the EEG signals perform dominant in the motor imagery tasks. In the following, the spatio-frequency information is extracted from the raw EEG data.

B. Data Processing

To sufficiently hold frequency feature of MI-EEG, two main bands of the EEG signals (i.e., μ: 8-13 Hz and β: 13-30 Hz rhythms) are further divided into three sub-bands which then correspond to three spatio-frequency images. The generation process of these images is shown in Fig. 1.

As shown in Fig. 1(a), CH_{64} represents the EEG signals of the 64-th electrodes in BCI2000 dataset. Since the raw EEG signal is non-stationary, the corresponding raw data is divided into N segments before operating FFT. The corresponding segments set of i-th electrode is given as

$$x_i = \{x_{i1}, x_{i2}, \dots, x_{iN}\} \tag{1}$$

where each segment x_{ij} is with l_j samples, $j = 1, 2, \dots, N$, and $x_{ij} \in \mathbb{R}^{l_j}$.

Next, FFT is utilized to transform each segment from time domain to frequency domain. Let $X_{ij} = \mathrm{FFT}[x_{ij}]$ which is divided into three components as following:

$$X_{ij} = \{X_{ij}^1, X_{ij}^2, X_{ij}^3\} \tag{2}$$

where $X_{ij}^m \in \mathbb{C}^{q_{ij}^m}$, $m = 1, 2, 3$ denote three sub-bands: 8-13Hz, 13-21Hz and 21-30Hz, respectively. q_{ij}^m is the length of X_{ij}^m.

Hence, for i-th electrode, the average power of m-th sub-band is calculated as:

$$P_i^m = \frac{1}{N} \sum_j X_{ij}^{mT} X_{ij}^m, m = 1, 2, 3, \tag{3}$$

Finally, we obtain the average power of each sub-band for all electrodes as shown in Fig. 1(b). Fig. 1(c) gives the 3-D space information of the 64-electrodes in practical experiment. In the next, to keep the space information of each electrode, azimuthal equidistant projection (AEP), which keeps each electrode position equidistant between 3-D and 2-D space, is adopted to convert the 3-D location of each electrode into 2-D position (the detailed location image as shown in Fig. 1(d)). Then, based on 2-D position of all electrodes, CT algorithm is employed to generate a two-dimensional image for each sub-band, and thus we obtain three two-dimensional spatio-frequency images corresponding to three sub-bands (as shown in Fig. 1(e)), each of which is an input of the proposed pDCNN. And, the detail of the structure of pDCNN will be illustrated in the next.

C. The structure of pDCNN

The overall structure of the proposed network model is shown in Fig. 2. The whole network structure mainly includes four parts: Input data, feature extraction, feature fusion and output. The input data process is shown in section II. This section will detail another three parts. In feature extraction part, three DCNN structures are adopted to correspond to images of three sub-bands. Each DCNN structure consists of four CNN blocks, each of which includes convolution layer, batch normalization layer and down sampling layer. Convolution layer is mainly used to extract feature and down

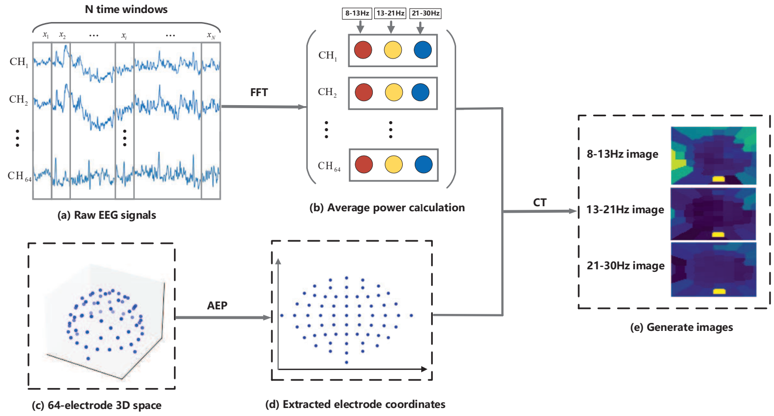


Fig. 1. Data processing

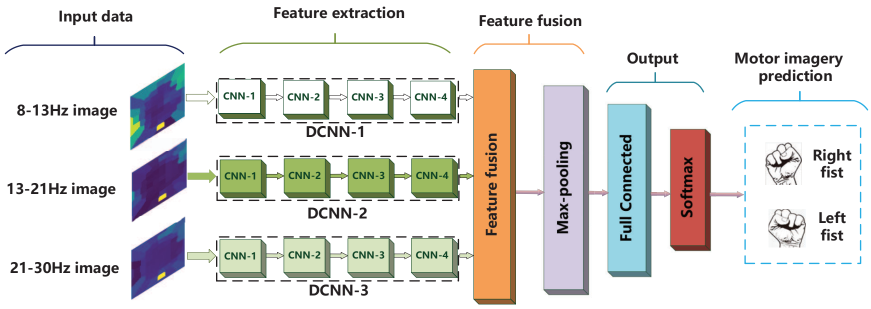


Fig. 2. The proposed pDCNN structure

sampling layer is used to reduce dimensionality. Besides, batch normalization layer is set to minimize covariate shift and enhance the robustness of the model, so as to overcome the overfitting problem. After extracting three groups features from three sub-bands (i.e., three images), these features are fused in feature fusion part. Then, a max-pooling block in feature fusion part is adopted to reduce dimensionality of the fusion feature. A full connected layer and softmax layer in the output are used to classify. Compared with the existing works fusing the spatio-frequency information of three sub-bands into a three-channel image, the proposed network directly deals with the single image corresponding to each sub-band, which can extract more features from each sub-band.

The detailed configuration is presented in Table 1. The proposed pDCNN structure consists of three DCNNs (i.e. DCNN-i, $i = 1, 2, 3$ in Table 1), each of which includes four blocks labeled as 1st, 2nd, 3rd, 4th. Besides, "BN" represents the batch normalization layer, and, the down sampling layer is denoted as "DS". In each convolution block, (conv3-32)$\times$2, for instance, means that there are 2 convolution layers with the size of the kernel is 3×3 and the total number of the kernels is 32. "ReLU" represents the rectified linear unit function, and "Max-pool" denotes the max-pooling layer. And, "FF" denotes the feature fusion part, which fuses three features from three DCNN parts. As shown in Table 1, the same configurations are contained for three DCNN which corresponds to three images, respectively. However, the parameters of three DCNN are irrelevant and only depend on their own inputs, respectively.

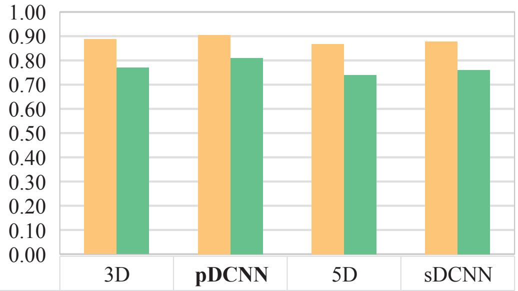
Table 1: The proposed pDCNN configuration

section:	DCNN–i
Input:	$64 \times 64 \times 1$ image
1^{st}:	(conv3-32) $\times$ 2
BN:	ReLU
DS:	Max-pool
2^{nd}:	(conv3-64) $\times$ 2
BN:	ReLU
DS:	Max-pool
3^{rd}:	(conv3-128) $\times$ 1
BN:	ReLU
DS:	Max-pool
4^{th}:	(conv3-256) $\times$ 1
BN:	ReLU
DS:	Max-pool
FF:	Feature Fusion Layer Max-pooling layer
Output:	Full connected layer Softmax

III. Experiments and Results

A. Experiments Parameters

1) Experimental Environment: The whole experiment is based on TensorFlow 1.8 environment, and runs on NVDIA



	3D	**pDCNN**	5D	sDCNN
Accuracy	0.8874	**0.9052**	0.8681	0.8780
Kappa Value	0.77	**0.81**	0.74	0.76

Fig. 3. The performance of different network structures.

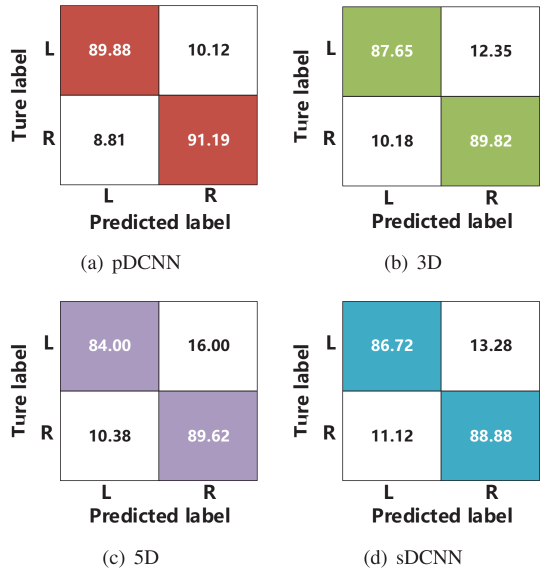


Fig. 4. Confusion matrices under different network models.

GTX1080Ti and Intel 2.1Ghz Xeon Silver 4110 CPU with 64G RAM.

2) Network Parameters: The detailed network parameters are set according to Table 1. And, the cross-entropy function is used as the loss function, and Adam update rule is employed as the optimizer to train the network model. The learning rate and dropout rate are chosen as 10^{-4} and 0.5, respectively.

3) Performance Indexes: Two evaluation criterions: Accuracy and Kappa value, are used to quantify the model classification performance, which are calculated by Eqs. (4) and (5), respectively.

$$Accuracy = \frac{TP}{TP + FN} \tag{4}$$

$$Kappa = \frac{Accuracy - Re}{1 - Re} \tag{5}$$

Where TP and FN represent the true posotives and false negatives fields in the confusion matrix, Re is 0.5 in two classification problems which represents the random classification rate.

B. Performance of different network structures

To show the performance of the proposed DCNN structures, four comparison experiments are conducted which include the proposed method and another three network models (i.e., 3D, 5D and sDCNN). All four comparison experiments are based on BCI2000 datasets. The training and test sets were combined and evaluated using 10-fold cross-validation (CV). The detailed configurations of another three network models are shown in Table 2. The "3D" and "5D" models are based on the proposed model to reduce or add a CNN block, that is, each DCNN in these structures contains three CNN blocks (i.e. 1st, 2nd, 3rd) and five CNN blocks (i.e. 1st, 2nd, 3rd, 4th, 5th), respectively. And, compared with above two models, the sDCNN model is a single DCNN structure, where three feature images are fused into a three-channel image as the input of this network structure. Besides, feature fusion layer and max-pool layer, which are contained in the "3D" and "5D" model, are removed in the sDCNN model. The classification functions of all network models adopt softmax function.

Fig. 3 intuitively shows the corresponding performance of the proposed method compared with three models. From Fig. 3, we can see that the Accuracy and Kappa value of the "3D" network structure are 88.74% and 0.77, respectively. However, the Accuracy and Kappa value of the "5D" network structure reduce to 86.81% and 0.74, respectively. Besides, for the sDCNN network structure, the corresponding Accuracy and Kappa value achieve at 87.80% and 0.76, respectively. And then, it is clear from Fig. 3 that the Accuracy of the proposed method reaches 90.52%, which reaches the apparent improvements of accuracy 1.78%, 3.71% and 2.72% compared with "3D", "5D" and sDCNN, respectively. And, the corresponding Kappa value increases to 0.81, which is larger than that of another three network structures. These results show that the proposed method achieves the best performance in classification of MI-EEG.

Next, Fig. 4 gives the confusion matrices under the different network structures to show the corresponding performance in MI-EEG classification tasks. As shown in Fig. 4, "L" and "R" represent two MI tasks: imaging left fist and imaging right fist, respectively. The accuracies of two categories are 89.88% and 91.19% as shown in Fig. 4(a), which represents the classification performance of the proposed pDCNN structure. Fig. 4(b) and Fig. 4(c) illustrate the confusion matrix of the "3D" and "5D" model, which shows the recognition accuracies of "L" and "R", respectively. In addition, the sDCNN model result in 86.72% of "L" and 88.88% of "R" classification accuracy, as the confusion matrix shown in Fig. 4(d). The accuracies of the proposed method are larger than the another three network structures, which shows that the proposed method achieves the most excellent classification result compared with another these structures. Further, from the above results, it can be seen that for the proposed method, the accuracy difference between two MI tasks is 1.31%, while the "3D", "5D" and

Table 2: Different DCNN Configurations

Model:	3D	5D	sDCNN
Input:	$3 \times 64 \times 64 \times 1$	$3 \times 64 \times 64 \times 1$	$64 \times 64 \times 3$
	DCNN–i	DCNN–i	—
1^{st}:	(conv3-32) $\times$ 2	(conv3-32) $\times$ 2	(conv3-32) $\times$ 2
BN:	ReLU	ReLU	ReLU
DS:	Max-pool	Max-pool	Max-pool
2^{nd}:	(conv3-64) $\times$ 2	(conv3-64) $\times$ 2	(conv3-64) $\times$ 2
BN:	ReLU	ReLU	ReLU
DS:	Max-pool	Max-pool	Max-pool
3^{rd}:	(conv3-128) $\times$ 1	(conv3-128) $\times$ 1	(conv3-128) $\times$ 1
BN:	ReLU	ReLU	ReLU
DS:	Max-pool	Max-pool	Max-pool
4^{th}:	—	(conv3-256) $\times$ 1	(conv3-256) $\times$ 1
BN:	—	ReLU	ReLU
DS:	—	Max-pool	Max-pool
5^{th}:	—	(conv3-512) $\times$ 1	—
BN:	—	ReLU	—
DS:	—	Max-pool	—
	Feature Fusion Layer	Feature Fusion Layer	—
	Max-pool Layer	Max-pool Layer	—
FC:	FC	FC	FC
Output:	softmax	softmax	softmax

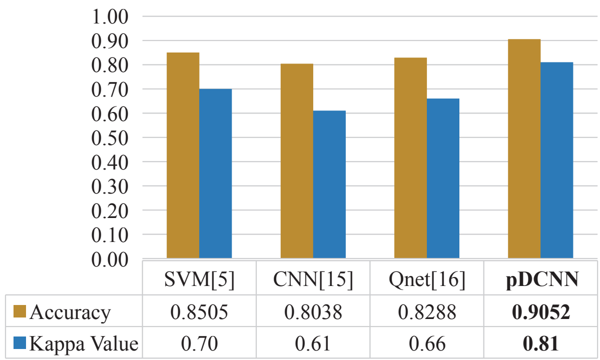


Fig. 5. The performance of different methods.

sDCNN correspond to differences 2.17%, 5.62% and 2.16%, respectively. These differences clearly indicates the accuracy gap of the proposed method is very smallest among all four comparison experiments, which shows the proposed method performs the most stable performance in consistency. Besides, by comparing the experiment results of the proposed method and the sDCNN model, the accuracies of the proposed method reach significant improvement, which reveals that the feature fusion process after extracting features for each sub-band leads to outstanding classification performance compared with directly fuse these as a three-channel image before inputting into network model.

C. Comparison of different methods

To highlight the excellent performance of the proposed method, the experiment results of the existing MI-EEG classification methods are presented as a comparison. The BCI2000 dataset is adopted in these methods and also employed in our work. The corresponding performance indexes are denoted by the average accuracy and Kappa value. The detailed results are shown in Fig. 5.

As can be seen from the figure ref Fig5, the SVM algorithm used in cite 5 has reached the accuracy of 85.05%and the kappa value of 0.70. And, the simple end-to-end CNN was adopted to directly extract features and achieve classification from raw EEG signals in [15] and the corresponding accuracy and Kappa value achieve at 80.38% and 0.61, respectively. In [16], QNet was employed to fuse bilinear vectors, which resulted in an accuracy of 82.88%and a kappa value of 0.66. However, pDCNN method increases to 90.52% and 0.81 in the accuracy and the Kappa value, which performs apparent improvement compared with these existing works.

IV. Conclusion

This paper investigates the decoding of MI-EEG tasks based on pDCNN. FFT is applied to obtain frequency information of the raw EEG signals, and the average power values of three sub-bands are extracted. Then, three corresponding MI-EEG images as the input of the classification model are generated by using the CT interpolation algorithm. Based on these MI-EEG images, a novel DCNN structure is designed to complete feature extraction and recognition of MI-EEG tasks. Compared with directly fusing features to a three-channel image before being fed into the classifier, the proposed method first extracts three groups features from three sub-bands by adopting three parallel DCNNs, respectively, and fuses these features in the fusion layer. Through the comparison with the existing related methods, the proposed method holds a significant improvement in accuracy and Kappa value. In future work, we will further develop a new data process method to deal with the non-stationary series so as to get good accuracy in MI-EEG classification tasks.

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